UC SAN DIEGO / ECE 251C

8

Wavelet processing and perfect-reconstruction filter banks

Executive summary

A wavelet processing method acquires fine-scale coefficients, applies a DWT, modifies the resulting coefficients, and reconstructs a signal through the inverse transform. We distinguish the acquisition approximation from the invertibility of the transform, and derive synthesis as upsampling followed by filtering. We then analyze a general two-channel bank in the z-domain. Perfect reconstruction requires the desired signal to pass with only a fixed delay and the alias contributions from the two branches to cancel. The Haar filters satisfy both conditions and preserve energy, giving a concrete connection between orthogonal wavelet expansions and digital filter banks.

8.1 From a wavelet representation to a processing method

Lecture 7 derived an analysis procedure that computes approximation and detail coefficients by filtering and downsampling. To use this representation for signal processing, we must also reconstruct from its coefficients. We begin by placing analysis and synthesis in the complete processing procedure, then derive the inverse operation and its reconstruction conditions. The processing itself takes place between the two transforms.

Recall the continuous-domain representation. Let ϕ and ψ generate an orthonormal wavelet system, with

$$\phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n), \qquad \psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n).$$

The approximation and detail coefficients are

$$a_i[n] = \langle f, \phi_{i,n}\rangle_{L^2}, \qquad d_i[n] = \langle f, \psi_{i,n}\rangle_{L^2}.$$

Keeping resolution zero as a reference gives

$$f(t) = \sum_{n\in\mathbb{Z}} a_0[n]\phi_{0,n}(t) + \sum_{i=0}^{\infty}\sum_{n\in\mathbb{Z}} d_i[n]\psi_{i,n}(t). \tag{8.1}$$

This is an equality in $L^2(\mathbb{R})$. The first term supplies the coarse approximation; each later level adds detail at a finer resolution.

The two-scale coefficients are

$$g_0[n] = \langle \phi, \phi_{1,n} \rangle_{L^2}, \qquad g_1[n] = \langle \psi, \phi_{1,n} \rangle_{L^2}. \tag{8.2}$$

These are discrete filters. We use real-valued ϕ, ψ and real filters below, as in the Haar example. The inner product still conjugates its second argument; that conjugation has no effect on these real functions.

8.2 The general wavelet analysis pipeline

The continuous-domain signal $f \in L^2(\mathbb{R})$ is not directly available to a digital algorithm. An acquisition device supplies measurements, such as pixel values or audio samples, that we organize into a sequence $a_I[n]$. Here I is the finest available resolution. Fix an orthonormal MRA with scaling function ϕ and wavelet ψ so that this sequence can be interpreted as coordinates of a continuous-domain approximation.

The fundamental modeling assumption is

$$\boxed{f(t) \approx f_I(t) \coloneqq \sum_{n \in \mathbb{Z}} a_I[n] \phi_{I,n}(t).} \tag{8.3}$$

In the ideal coefficient model, $a_I[n] = \langle f, \phi_{I,n} \rangle_{L^2}$ and $f_I = P_I f$. For a physical device, the identification of its measurements with these coefficients must be justified by the acquisition model. This separates two questions: how accurately f_I represents f, and whether a transform can recover a_I from its transformed coefficients. Perfect reconstruction addresses the second question, not the first.

Choose a positive integer K for the number of decomposition levels. The resulting processing method consists of analysis, coefficient modification, inverse analysis, and continuous-domain reconstruction. We use bars to distinguish processed coefficients and signals from their original values.

Step 1: perform a K-level DWT. At each level, apply the low-pass and high-pass analysis filters and downsample both outputs by two. Continue with only the approximation output, retaining each detail sequence at its own level:

$$a_I \xrightarrow{\text{DWT}} (a_{I-K}, d_{I-K}, \ldots, d_{I-1}).$$

Thus there is one coarse approximation and K detail sequences. For example, three levels give $(a_{I-3}, d_{I-3}, d_{I-2}, d_{I-1})$, not just the coarsest approximation a_{I-3}.

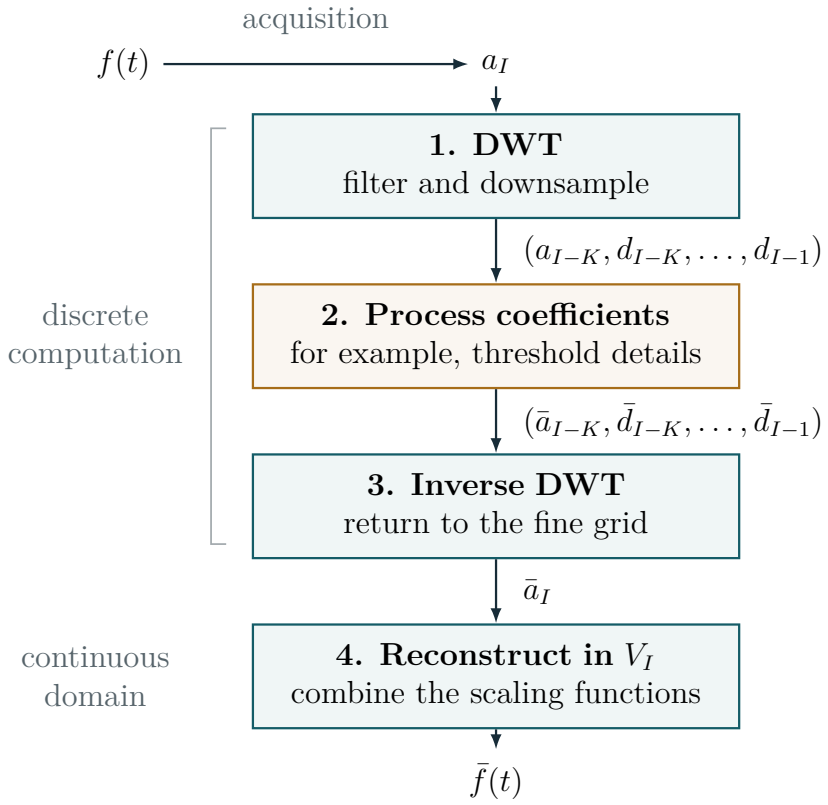


Figure 8.1: The complete pipeline. Acquisition supplies a_I; the DWT reorganizes it by scale, processing changes the coefficients, and the inverse DWT produces $\bar{a}_I$. The last step represents that output as a function $\bar{f} \in V_I$.

Step 2: process the wavelet coefficients. Modify this collection to obtain

$$(\bar{a}_{I-K}, \bar{d}_{I-K}, \ldots, \bar{d}_{I-1}).$$

A common choice is to leave a_{I-K} unchanged and threshold only the details. More generally, the processing rule can act on the entire collection. A bar indicates the output of this processing; it does not denote complex conjugation, for which we continue to use $*$. Hats remain reserved for Fourier transforms.

Step 3: perform the inverse DWT. Starting from the processed coarse approximation, reintroduce the details from coarsest to finest:

$$(\bar{a}_{I-K}, \bar{d}_{I-K}, \ldots, \bar{d}_{I-1}) \xrightarrow{\text{IDWT}} \bar{a}_I.$$

The first inverse stage combines $\bar{a}_{I-K}$ with $\bar{d}_{I-K}$ to form $\bar{a}_{I-K+1}$. Each subsequent stage incorporates the detail at the next finer resolution, until resolution I is reached. The output represents the modified coordinates. In particular, invertibility of the transform does not imply invertibility of the intervening processing rule.

Step 4: form the processed continuous-domain signal. Use the same fine-resolution scaling functions as in the acquisition model:

$$\boxed{\bar{f}(t) \coloneqq \sum_{n\in\mathbb{Z}} \bar{a}_I[n]\phi_{I,n}(t).} \tag{8.4}$$

The inverse DWT returns a sequence of fine-scale coefficients. The expansion above identifies the function represented by that sequence. Thus f_I and $\bar{f}$ belong to the same approximation space V_I; processing changes their coordinates, not the underlying space.

Without coefficient modification. An exact inverse DWT returns $\bar{a}_I = a_I$, so $\bar{f} = f_I$. Equality with the underlying f additionally requires the acquisition model to be exact. The rest of the lecture explains how filtering and rate changes implement these forward and inverse operations.

8.3 The multilevel discrete wavelet transform

At one level, the analysis filters are the reversed two-scale filters:

$$h_0[n] = g_0[-n], \qquad h_1[n] = g_1[-n].$$

For complex filters the corresponding formula is $h_\ell[n] = g_\ell^*[-n]$; the real-valued setting reduces it to reversal. Writing $\mathrm{H}_\ell x = h_\ell * x$, the two outputs are

$$\boxed{a_i[m] = (h_0 * a_{i+1})[2m], \qquad d_i[m] = (h_1 * a_{i+1})[2m].} \tag{8.5}$$

The symbol $\downarrow 2$ means *retain the even-indexed entries*: $(\downarrow 2\, v)[m] = v[2m]$. Filtering combines neighboring coefficients; downsampling then expresses each output on the coarser grid.

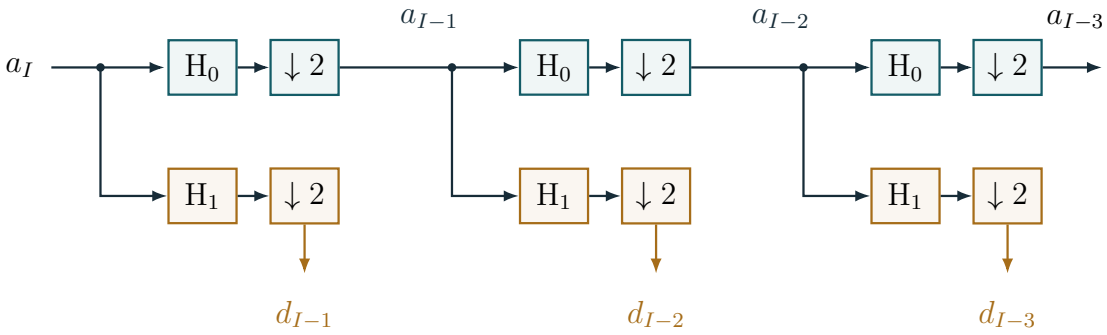


Figure 8.2: A three-level DWT. Only the approximation output is split again. Each detail output is retained at the resolution where it was computed.

The first split replaces a_I by (a_{I-1}, d_{I-1}). The second replaces only a_{I-1}, and the third replaces only a_{I-2}. Thus the final stored representation is

$$a_{I-3}, \quad d_{I-3}, \quad d_{I-2}, \quad d_{I-1}.$$

These sequences together describe the same f_I:

$$f_I = \sum_n a_{I-3}[n]\phi_{I-3,n} + \sum_{i=I-3}^{I-1} \sum_n d_i[n]\psi_{i,n}. \tag{8.6}$$

The finest detail index is $I-1$, not I. For K decomposition levels, the coarsest retained approximation is a_{I-K}.

Although the representation is defined using ϕ and ψ, the algorithm needs only the associated filter coefficients. The basis functions determine what the coordinates mean, while the filter bank determines how those coordinates are computed.

8.4 Processing changes the coefficients, not the basis

An invertible change of coordinates does not by itself remove noise or reduce the information needed to represent a signal. Its value is that the new coordinates may separate important signal features from coefficients that can be reduced or discarded. In a wavelet representation, a common choice is to preserve the coarse approximation and threshold the details.

Example: hard thresholding. For a threshold $\tau > 0$, define

$$\eta_\tau(c) = \begin{cases} 0, & |c| \le \tau, \\ c, & |c| > \tau. \end{cases}$$

Small coefficients are set to zero; the others are retained. With K decomposition levels, the processed coefficients are

$$\bar{a}_{I-K} = a_{I-K}, \qquad \bar{d}_i[n] = \eta_\tau(d_i[n]), \quad I-K \le i < I.$$

For three levels, this acts on $d_{I-3}, d_{I-2}, d_{I-1}$. Choosing the threshold is a separate design question; here we are identifying where the processing enters.

Thresholding is nonlinear. For example, each of two coefficients equal to $3\tau/4$ is sent to zero, but their sum $3\tau/2$ is retained. Thus thresholding a sum is not the same as adding the thresholded values.

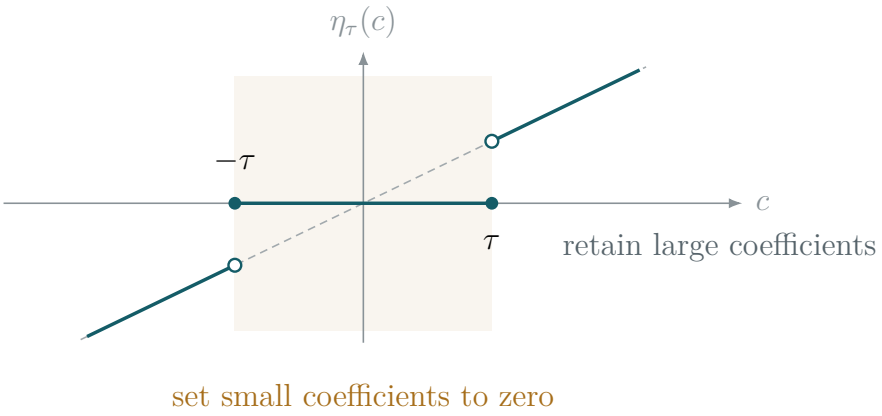


Figure 8.3: Hard thresholding for a real coefficient. The dashed line is the identity map. The jump at each threshold is part of the processing rule.

Think analog, act digital

The MRA interprets the signal and its scales; the computation uses only sequences and filters. After processing, the inverse DWT faithfully synthesizes the *modified* coordinates. It does not undo thresholding. The resulting $\bar{a}_I$ specifies $\bar{f}$ through equation (8.4).

8.5 Recovering the fine coefficients

We now construct the inverse operation used in the processing procedure. Since a multilevel DWT repeats the same decomposition, it is enough to invert one stage. The coarse approximation and its detail must recover the finer approximation:

$$f_{i+1} = \sum_m a_i[m]\phi_{i,m} + \sum_m d_i[m]\psi_{i,m}.$$

The two-scale equations express both families using the finer scaling functions. With the same masks as in equation (8.2),

$$\phi_{i,m} = \sum_n g_0[n-2m]\phi_{i+1,n},$$
$$\psi_{i,m} = \sum_n g_1[n-2m]\phi_{i+1,n}.$$

For example, substitute $2^i t - m$ into $\phi(t) = \sum_k g_0[k]\sqrt{2}\,\phi(2t-k)$ and multiply by $2^{i/2}$. The fine index is $n = 2m + k$, so the mask index is $k = n - 2m$.

Substituting both expansions and collecting the coefficient of $\phi_{i+1,n}$ gives

$$\begin{aligned} f_{i+1} &= \sum_m \sum_n a_i[m] g_0[n-2m] \phi_{i+1,n} \\ &\quad + \sum_m \sum_n d_i[m] g_1[n-2m] \phi_{i+1,n} \\ &= \sum_n \left(\sum_m a_i[m] g_0[n-2m] + \sum_m d_i[m] g_1[n-2m] \right) \phi_{i+1,n}. \end{aligned}$$

Uniqueness of the coordinates in this orthonormal basis yields

$$\boxed{a_{i+1}[n] = \sum_m a_i[m] g_0[n-2m] + \sum_m d_i[m] g_1[n-2m].} \tag{8.7}$$

Why upsampling appears. Define zero-insertion upsampling by

$$(\uparrow 2\, c)[n] := \begin{cases} c[n/2], & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases} \tag{8.8}$$

Then convolution with g satisfies

$$(g * (\uparrow 2\, c))[n] = \sum_k g[n-k] (\uparrow 2\, c)[k] = \sum_m g[n-2m] c[m].$$

Only even $k = 2m$ survive. Consequently equation (8.7) is exactly

$$a_{i+1} = g_0 * (\uparrow 2\, a_i) + g_1 * (\uparrow 2\, d_i). \tag{8.9}$$

A synthesis bank. The inverse stage first inserts zeros, then filters each channel, and finally adds the results. Upsampling places each coarse coefficient on the fine grid; the filters then distribute its contribution among the finer coefficients.

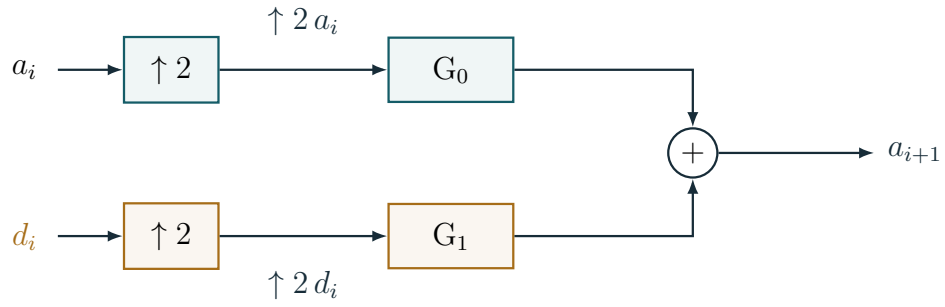


Figure 8.4: Synthesis reverses one analysis stage. Here $G_\ell c = g_\ell * c$. The order is *upsample, then filter*, as required by the index $n - 2m$ in the synthesis formula.

Zero insertion is not interpolation. Consider the three nonzero entries $c[0] = 1$, $c[1] = 2$, $c[2] = -1$. Upsampling changes their locations to $0, 2, 4$ and fills the odd locations with zeros. It does not infer missing values.

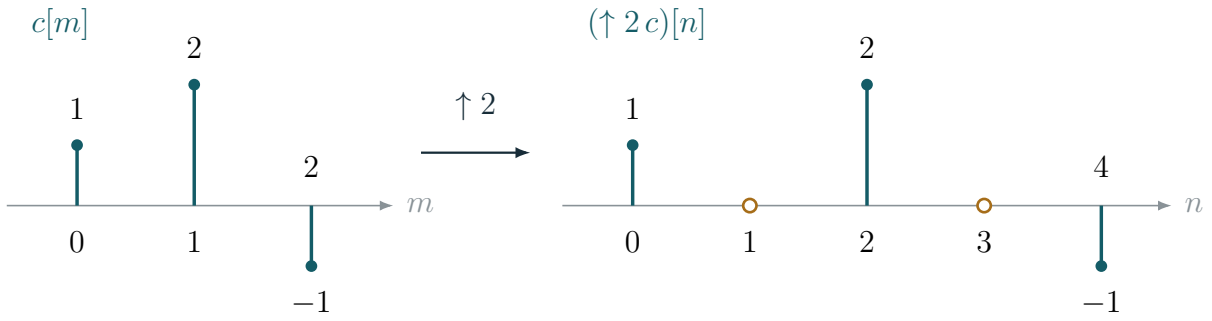


Figure 8.5: The open gold circles mark inserted zeros. Filtering after this step supplies the contributions needed on the fine grid.

The order follows from the synthesis formula: the mask is evaluated at $n - 2m$, where n is a fine-grid index and m is a coarse-grid index. Filtering first on the coarse grid and then inserting zeros would apply a different operator unless the filter were changed accordingly.

For processed coefficients, the same synthesis formula applies with bars on every a_i and d_i. A three-level inverse DWT first combines $\bar{a}_{I-3}$ and $\bar{d}_{I-3}$ to produce $\bar{a}_{I-2}$, then uses $\bar{d}_{I-2}$, and finally $\bar{d}_{I-1}$. The result is the processed sequence $\bar{a}_I$.

8.6 Downsampling followed by upsampling

To derive the reconstruction conditions, first examine the downsampler and upsampler without the filters. Their composition acts on the original index grid, which allows its effect to be expressed by a simple pointwise mask. Let

$$c[m] = v[2m], \qquad w = \uparrow 2\, c.$$

Then $w[2m] = v[2m]$ and $w[2m+1] = 0$. The cascade returns the sequence to its original index grid, but it does not restore the discarded odd entries.

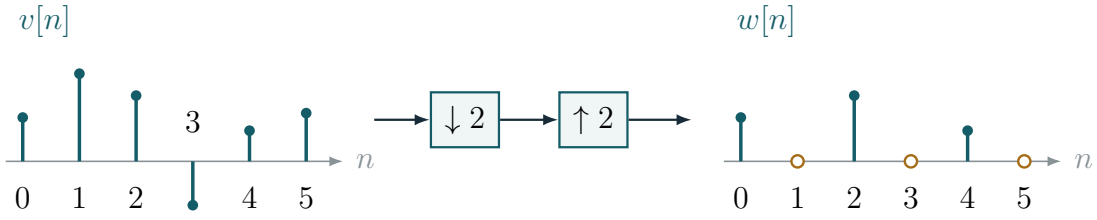


Figure 8.6: Only the even entries survive. The two rate changes together act as an alternating keep/erase mask on the original grid.

The even-index mask has a useful algebraic expression:

$$\frac{1+(-1)^n}{2} = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Therefore

$$\boxed{w[n] = \frac{1}{2}\big(v[n] + (-1)^n v[n]\big).} \tag{8.10}$$

To transform the second term, recall the bilateral z-transform $V(z) = \sum_n v[n] z^{-n}$. Since $(-1)^{-n} = (-1)^n$ for integers n,

$$\sum_n (-1)^n v[n] z^{-n} = \sum_n v[n](-z)^{-n} = V(-z).$$

Thus the combined rate changes give

$$\boxed{W(z) = \frac{1}{2}\big(V(z) + V(-z)\big).} \tag{8.11}$$

On the unit circle, $-\mathrm{e}^{\mathrm{j}\omega} = \mathrm{e}^{\mathrm{j}(\omega+\pi)}$, so $V(-z)$ is a copy of the spectrum shifted by π. This term is created by the removal of the odd-indexed samples. It will appear in both branches of the bank; reconstruction requires their contributions to cancel after synthesis.

8.7 The two-channel reconstruction formula

We can now examine the analysis and synthesis banks together. The input x is an arbitrary discrete sequence; in the wavelet algorithm it is one of the approximation sequences a_{i+1}. There is no coefficient modification in the following diagram.

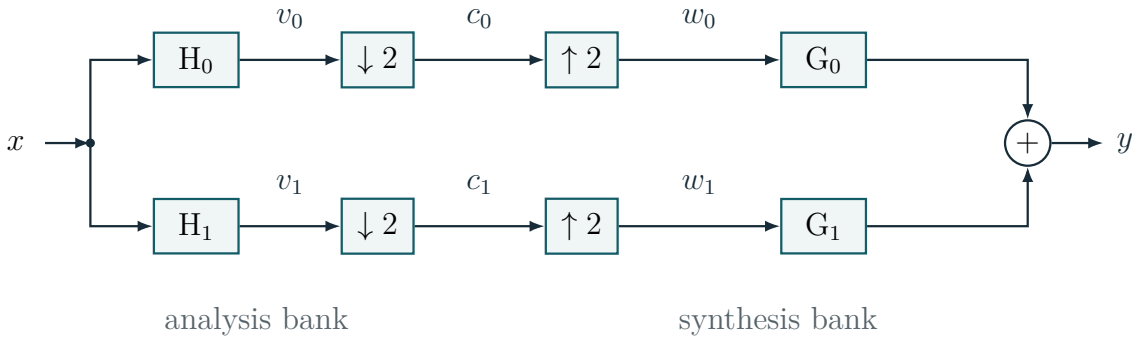


Figure 8.7: Each analysis branch filters and downsamples. Each synthesis branch upsamples and filters. Both branch outputs are added on the original index grid.

For the algebra below, take the filters to be FIR and begin with a finite input, so every transform is a finite sum of powers of z. In branch $\ell \in \{0, 1\}$,

$$V_\ell(z) = H_\ell(z) X(z).$$

Applying equation (8.11), then the synthesis filter, gives the branch contribution

$$G_\ell(z)W_\ell(z) = \frac{1}{2}G_\ell(z)\big(H_\ell(z)X(z) + H_\ell(-z)X(-z)\big).$$

Adding the two branches and grouping the two input terms yields

$$\begin{aligned} Y(z) &= \frac{1}{2}G_0(z)\big(H_0(z)X(z) + H_0(-z)X(-z)\big) \\ &\quad + \frac{1}{2}G_1(z)\big(H_1(z)X(z) + H_1(-z)X(-z)\big) \\ &= \underbrace{\frac{G_0(z)H_0(z) + G_1(z)H_1(z)}{2}}_{T(z)} X(z) \\ &\quad + \underbrace{\frac{G_0(z)H_0(-z) + G_1(z)H_1(-z)}{2}}_{A(z)} X(-z). \end{aligned} \tag{8.12}$$

The response $T(z)$ multiplies the original spectrum and determines its gain and phase. The response $A(z)$ multiplies the shifted spectrum and determines the alias contribution. Thus a branch containing rate changes is not generally described by one convolution response on the input grid. The sum of the two branches becomes such a response when the alias term vanishes.

8.8 Two conditions for perfect reconstruction

Definition. The filter bank has *perfect reconstruction* (PR) if, without coefficient processing,

$$y[n] = x[n - L] \qquad \text{for every input } x,$$

for a fixed integer L. A nonnegative L is a delay; $L = 0$ gives exact sample-by-sample reconstruction. Its transform is $Y(z) = z^{-L}X(z)$: all frequencies must have unit gain and the same time delay.

The following conditions are associated with Vetterli's 1986 work on perfect-reconstruction filter banks. They separate the reconstruction requirement into the two identities suggested by equation (8.12).

Proposition 8.1. For the two-channel FIR bank above, perfect reconstruction with shift L is equivalent to

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2z^{-L}, \qquad \text{no distortion,} \tag{8.13}$$

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0, \qquad \text{no aliasing.} \tag{8.14}$$

The factor of two in the no-distortion condition is essential: it compensates for the factor 1/2 introduced by the even-index selector. The second identity removes the shifted copy rather than attempting to reconstruct the input from either downsampled branch alone.

Proof. If both identities hold, equation (8.12) reduces to $Y(z) = z^{-L}X(z)$.

For necessity, the reconstruction rule must hold for every input. With $x[n] = \delta[n]$, we have $X(z) = X(-z) = 1$, so

$$T(z) + A(z) = z^{-L}.$$

With $x[n] = \delta[n-1]$, we have $X(z) = z^{-1}$ and $X(-z) = -z^{-1}$. Hence

$$T(z) - A(z) = z^{-L}.$$

Adding and subtracting gives $T(z) = z^{-L}$ and $A(z) = 0$, which are exactly the two stated identities. □

Why alias cancellation is possible. Each downsampled branch can contain aliasing. Perfect reconstruction does not require each branch to reconstruct the input separately. It requires their *sum* to cancel the alias terms while retaining the desired signal.

Why a delay is allowed. A delay changes only the time origin. It does not alter relative amplitudes or waveform shape. A frequency-dependent gain, by contrast, would change the signal even after the output is shifted back.

8.9 The Haar analysis and synthesis filters

The general conditions can be checked directly for the Haar bank. We keep the same tap alignment used to derive its projection formulas, so that the algebra and the coefficient calculation describe the same transform. The synthesis filters have their nonzero taps at 0 and 1:

$$g_0[n] = \frac{\delta[n] + \delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n] - \delta[n-1]}{\sqrt{2}}.$$

Reversing these sequences places the analysis taps at -1 and 0:

$$h_0[n] = \frac{\delta[n+1] + \delta[n]}{\sqrt{2}}, \qquad h_1[n] = \frac{\delta[n] - \delta[n+1]}{\sqrt{2}}.$$

This is the same alignment used in the projection calculation: analysis combines $x[2m]$ with $x[2m+1]$. The negative tap index makes that convention explicit. It also allows the complete analysis–synthesis bank to reconstruct with zero delay.

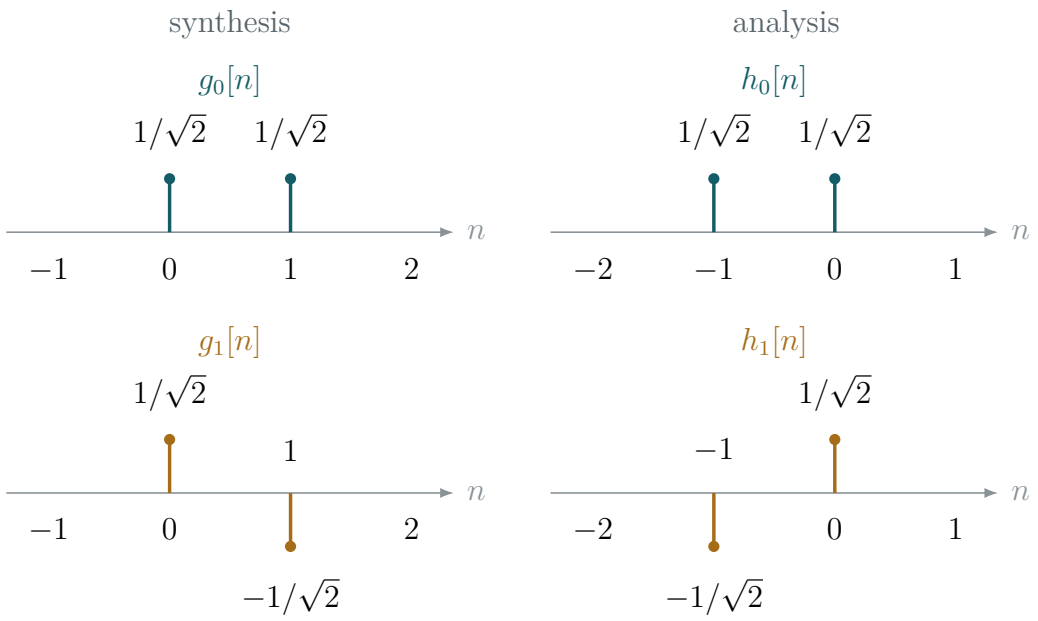


Figure 8.8: Reversal changes the tap locations, not their values. The high-pass filters have opposite signs at neighboring indices.

Using $\delta[n-k] \xleftrightarrow{\mathcal{Z}} z^{-k}$, we obtain

$$\begin{aligned} G_0(z) &= \frac{1+z^{-1}}{\sqrt{2}}, \quad H_0(z) = \frac{1+z}{\sqrt{2}}, \\ G_1(z) &= \frac{1-z^{-1}}{\sqrt{2}}, \quad H_1(z) = \frac{1-z}{\sqrt{2}}. \end{aligned} \tag{8.15}$$

In particular, the tap at $n=-1$ contributes z, not z^{-1}. We now substitute these four expressions into both PR identities.

No distortion. Expand each product before adding:

$$\begin{aligned} G_0(z)H_0(z) + G_1(z)H_1(z) &= \frac{1}{2}(1+z^{-1})(1+z) + \frac{1}{2}(1-z^{-1})(1-z) \\ &= \frac{1}{2}\big(2+z+z^{-1}\big) + \frac{1}{2}\big(2-z-z^{-1}\big) \\ &= 2. \end{aligned}$$

Thus equation (8.13) holds with $L=0$. This zero-delay alignment uses the analysis tap at -1; it is not a claim that both analysis filters are causal.

No aliasing. First substitute $-z$ into the analysis responses:

$$H_0(-z) = \frac{1-z}{\sqrt{2}}, \qquad H_1(-z) = \frac{1+z}{\sqrt{2}}.$$

Then

$$\begin{aligned} G_0(z)H_0(-z) + G_1(z)H_1(-z) &= \frac{1}{2}(1+z^{-1})(1-z) + \frac{1}{2}(1-z^{-1})(1+z) \\ &= \frac{1}{2}\big(z^{-1} - z\big) + \frac{1}{2}\big(z - z^{-1}\big) \\ &= 0. \end{aligned}$$

The two alias contributions are nonzero in general, but are exact opposites. Combining both calculations in equation (8.12) gives $Y(z) = X(z)$.

Seeing the cancellation on the unit circle. Write

$$T_\ell(z) = \tfrac{1}{2}G_\ell(z)H_\ell(z), \qquad A_\ell(z) = \tfrac{1}{2}G_\ell(z)H_\ell(-z).$$

At $z = \mathrm{e}^{\mathrm{j}\omega}$, the identities $z + z^{-1} = 2\cos\omega$ and $z - z^{-1} = 2\mathrm{j}\sin\omega$ give

$$\begin{aligned} T_0 &= \cos^2(\omega/2), \quad T_1 = \sin^2(\omega/2), \\ A_0 &= -\tfrac{\mathrm{j}}{2}\sin\omega, \quad A_1 = \tfrac{\mathrm{j}}{2}\sin\omega. \end{aligned}$$

The first two responses add to one; the last two add to zero.

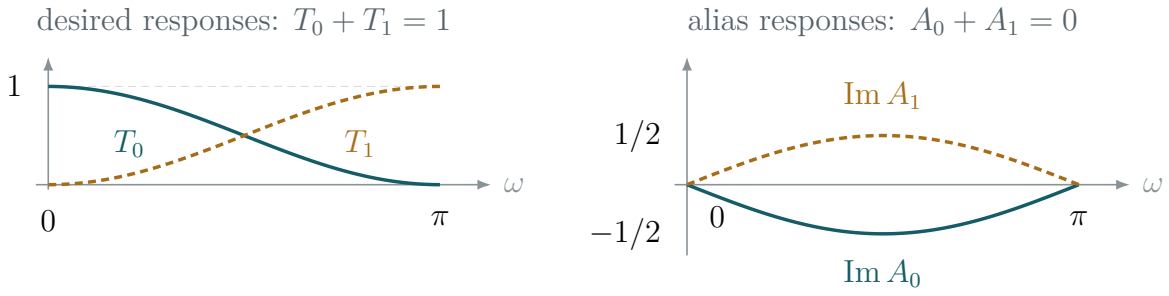


Figure 8.9: For the Haar bank, the desired parts complement one another while the imaginary alias parts cancel. The plots show branch contributions *after* synthesis, including their factor 1/2.

8.10 Perfect reconstruction on each pair of samples

The z-domain calculation proves reconstruction for every input. For Haar, we can also see the reconstruction directly on each pair of coefficients, which makes the role of the two channels explicit. A direct check uses $(h_\ell * x)[2m] = \sum_k h_\ell[k]x[2m-k]$. With only the taps at $k = -1, 0$ nonzero, analysis becomes

$$a[m] = \frac{x[2m] + x[2m+1]}{\sqrt{2}}, \qquad d[m] = \frac{x[2m] - x[2m+1]}{\sqrt{2}}. \tag{8.16}$$

Let $u = \uparrow 2\, a$ and $v = \uparrow 2\, d$. Synthesis is

$$y[n] = \frac{u[n] + u[n-1]}{\sqrt{2}} + \frac{v[n] - v[n-1]}{\sqrt{2}}.$$

For $n = 2m$, the entries $u[n-1]$ and $v[n-1]$ vanish. For $n = 2m+1$, the entries $u[n]$ and $v[n]$ vanish. Consequently

$$y[2m] = \frac{a[m] + d[m]}{\sqrt{2}} = \frac{2x[2m]}{2} = x[2m],$$
$$y[2m+1] = \frac{a[m] - d[m]}{\sqrt{2}} = \frac{2x[2m+1]}{2} = x[2m+1].$$

In the sum, the odd sample cancels; in the difference, the even sample cancels. Both entries are recovered. Retaining both channels, rather than either channel alone, preserves the pair.

The Haar bank is orthogonal. In matrix form,

$$\begin{pmatrix} a[m] \\ d[m] \end{pmatrix} = U \begin{pmatrix} x[2m] \\ x[2m+1] \end{pmatrix}, \qquad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$U^{\mathsf{T}} U = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I_2.$$

The inverse is $U^{\mathsf{T}} = U$, and $|a[m]|^2 + |d[m]|^2 = |x[2m]|^2 + |x[2m+1]|^2$. Summing over m shows that the complete analysis bank preserves sequence energy:

$$\|a\|_{\ell^2}^2 + \|d\|_{\ell^2}^2 = \|x\|_{\ell^2}^2.$$

The two-channel transform therefore preserves both the samples and their total energy. This is the coefficient-domain counterpart of the orthogonal decomposition $V_{i+1} = V_i \oplus W_i$. General perfect-reconstruction banks need not preserve energy; that additional property distinguishes orthogonal banks.

What perfect reconstruction guarantees

Without coefficient processing, synthesis returns exactly the acquired sequence. With processing, it returns the signal specified by the modified coefficients. Neither assertion removes the approximation made during acquisition.